

Pure-State π -Bases Obstruct the Negative Von Neumann Cases

Run fa_banach_001, lane 19

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. It proves that a von Neumann algebra has BCP, equivalently UBCP, if and only if it is atomic and separably represented. The same paper asks for a broader noncommutative C^* -algebra characterization:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

For a C^* -algebra A , let $\mathcal{P}(A)$ denote its pure state space, with the relative weak-star topology inherited from A^* .

Theorem 1. *Let M be a von Neumann algebra. If $\mathcal{P}(M)$ has a countable π -base, then M is atomic and separably represented. Consequently M has BCP and UBCP by the source paper's von Neumann classification.*

Corollary 1. *If a von Neumann algebra M fails BCP, then $\mathcal{P}(M)$ has no countable π -base.*

Idea of the Proof

The proof is by contrapositive and splits exactly along the source classification.

If M has an atomless direct summand, all pure states on that summand are singular. Given countably many proposed pure-state open sets, choose one pure state from each. A common-projection lemma for countably many singular states gives a nonzero projection p annihilating all of them. But the set of pure states satisfying $\psi(p) > 1/2$ is a nonempty open set, and none of the proposed open sets is contained in it.

If M is atomic but not separably represented, then either it has uncountably many atomic summands, or one summand is $\mathcal{B}(H)$ for a nonseparable Hilbert space H . In both cases one obtains uncountably many pairwise disjoint nonempty open subsets of the pure state space, which rules out a countable π -base.

Atomless Summands

Lemma 1 (common projection for singular pure states). *Let N be a nonzero von Neumann algebra and let (φ_n) be countably many singular states on N . Then there is a nonzero projection $p \in N$ such that*

$$\varphi_n(p) = 0 \quad (n \geq 1).$$

Proof. Fix a normal state ω on N . A singular positive functional φ admits an increasing net of projections $e_i \uparrow 1$ such that $\varphi(e_i) = 0$. Since ω is normal, for every $\varepsilon > 0$ there is a projection e with

$$\varphi(e) = 0, \quad \omega(1 - e) < \varepsilon.$$

Apply this to φ_n with $\varepsilon_n = 2^{-n-2}$, obtaining projections e_n with

$$\varphi_n(e_n) = 0, \quad \omega(1 - e_n) < 2^{-n-2}.$$

Let

$$p = \bigwedge_{n=1}^{\infty} e_n.$$

Then

$$\omega(1 - p) = \omega\left(\bigvee_n (1 - e_n)\right) \leq \sum_n \omega(1 - e_n) < \frac{1}{2}.$$

Thus $p \neq 0$. Since $p \leq e_n$, positivity gives $\varphi_n(p) = 0$ for every n . □

Lemma 2. *If N is a nonzero atomless von Neumann algebra, then $\mathcal{P}(N)$ has no countable π -base.*

Proof. In an atomless von Neumann algebra there are no normal pure states, because normal pure states are exactly vector states supported on minimal projections. Thus every pure state on N is singular.

Let (U_n) be any countable family of nonempty open subsets of $\mathcal{P}(N)$. Choose $\varphi_n \in U_n$. By Lemma 1, there is a nonzero projection $p \in N$ with $\varphi_n(p) = 0$ for every n .

Choose a pure state ψ_0 on the nonzero unital C^* -algebra pNp . Its canonical extension

$$\psi(x) = \psi_0(pxp) \quad (x \in N)$$

is a pure state of N , since pure states of hereditary subalgebras extend purely to the ambient algebra. Moreover $\psi(p) = 1$.

Therefore

$$W = \{\psi \in \mathcal{P}(N) : \psi(p) > 1/2\}$$

is a nonempty open subset of $\mathcal{P}(N)$. But $\varphi_n \in U_n \setminus W$ for every n . Hence no U_n is contained in W , so (U_n) cannot be a π -base. Since the countable family was arbitrary, $\mathcal{P}(N)$ has no countable π -base. □

Atomic Nonseparable Summands

Lemma 3. *If a topological space has uncountably many pairwise disjoint nonempty open sets, then it has no countable π -base.*

Proof. If (V_n) were a countable π -base and $(O_i)_{i \in I}$ were pairwise disjoint nonempty open sets, then for each i some $V_{n(i)}$ would satisfy $V_{n(i)} \subset O_i$. Distinct i 's require distinct $n(i)$'s, because the O_i 's are disjoint and $V_{n(i)} \neq \emptyset$. Thus I would be countable. $\square$

Lemma 4. *Let M be an atomic von Neumann algebra which is not separably represented. Then $\mathcal{P}(M)$ has no countable π -base.*

Proof. Write

$$M \cong \prod_{i \in I} \mathcal{B}(H_i).$$

If I is uncountable, let z_i be the central projection onto the i -th summand. The sets

$$O_i = \{\varphi \in \mathcal{P}(M) : \varphi(z_i) > 1/2\}$$

are pairwise disjoint nonempty open subsets of $\mathcal{P}(M)$. By Lemma 3, $\mathcal{P}(M)$ has no countable π -base.

If I is countable but M is not separably represented, then some summand is $\mathcal{B}(H)$ with H nonseparable. It is enough to show that $\mathcal{P}(\mathcal{B}(H))$ has no countable π -base, since this pure-state space is a clopen subspace of $\mathcal{P}(M)$.

Choose an uncountable orthonormal family $(\xi_\alpha)_{\alpha \in \Gamma}$ in H , and let p_α be the rank-one projection onto $\mathbb{C}\xi_\alpha$. The sets

$$W_\alpha = \{\varphi \in \mathcal{P}(\mathcal{B}(H)) : \varphi(p_\alpha) > 1/2\}$$

are nonempty because they contain the vector states ω_{ξ_α} . They are open. They are pairwise disjoint: if $\alpha \neq \beta$, then $p_\alpha + p_\beta \leq 1$, so no state can take value $> 1/2$ on both p_α and p_β . Lemma 3 applies again. $\square$

Proof of the Main Result

Proof of Theorem 1. We prove the contrapositive. Suppose M is not atomic and separably represented.

If M has a nonzero atomless central summand $N = zM$, then

$$\mathcal{P}(N) = \{\varphi \in \mathcal{P}(M) : \varphi(z) = 1\}$$

is a clopen subspace of $\mathcal{P}(M)$. By Lemma 2, $\mathcal{P}(N)$ has no countable π -base. Therefore $\mathcal{P}(M)$ has no countable π -base, because every open subspace of a space with a countable π -base inherits one.

If M is atomic but not separably represented, then Lemma 4 gives the same conclusion.

Thus $\mathcal{P}(M)$ having a countable π -base forces M to be atomic and separably represented. The source paper's von Neumann theorem then gives BCP and UBCP. $\square$

Proof of Corollary 1. The source paper proves that a von Neumann algebra fails BCP exactly when it is not atomic and separably represented. Apply Theorem 1. $\square$

Scope and Limitations

This packet proves a one-way compatibility theorem for the pure-state π -base route in the von Neumann class:

$$\mathcal{P}(M) \text{ has a countable } \pi\text{-base} \implies M \text{ has UBCP.}$$

It does not prove the converse for all atomic separably represented von Neumann algebras, and it does not settle arbitrary C^* -algebras. Its role is to show that the source's von Neumann negative cases are already detected at the level of pure-state π -bases.

Novelty and Search Bounds

Local indexes were searched for arXiv:2311.14257, von Neumann BCP, pure state space, π -base, singular pure state, atomless von Neumann algebra, and atomic nonseparable representation. The source paper contains the von Neumann BCP/UBCP classification, and previous lane-19 packets contain the countable-capture refinement. No prior local packet recorded this pure-state π -base obstruction for the negative von Neumann cases.

Human Review Focus

Reviewers should check:

1. the common-projection lemma for countably many singular states;
2. the assertion that atomless von Neumann algebras have no normal pure states;
3. the pure extension from pNp to N ;
4. the inheritance of countable π -bases by open subspaces;
5. the uncountable disjoint-open-family arguments in the atomic nonseparable cases.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Bruce Blackadar, *Operator Algebras: Theory of C^* -Algebras and Von Neumann Algebras*, Encyclopaedia of Mathematical Sciences 122, Springer, 2006.
- [3] Run `fa_banach_001`, lane 19, *Countable Capture for Von Neumann Algebras*, solution packet 2311.14257_von_neumann_capture_classification, 2026.